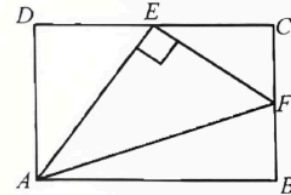


L11

Q1

- As shown in the figure below, in rectangle $ABCD$, E and F are two points on $\overline{CD}$ and $\overline{BC}$. If $m\angle AEF = 90^\circ$, then which of the following must be correct?



- A. $\triangle ADE \sim \triangle ECF$ B. $\triangle ECF \sim \triangle AEF$ C. $\triangle ADE \sim \triangle AEF$ D. $\triangle AEF \sim \triangle ABF$

1. **【 Answer 】** A

【 Solution 】 In rectangle $ABCD$, $m\angle D = m\angle C = 90^\circ$.

Since $m\angle AEF = 90^\circ$,

we have $m\angle DEA + m\angle CEF = 90^\circ$, $m\angle DEA + m\angle DAE = 90^\circ$,

thus $m\angle DAE = m\angle CEF$,

hence $\triangle ADE \sim \triangle ECF$.

Q2

- The number of correct statements among the following is ____ .

- ① all squares are similar
- ② all rhombi are similar
- ③ all rhombi with equal side lengths are similar
- ④ all rectangles with equal diagonal lengths are similar

2. **【 Answer 】** 1

【 Solution 】 ① is correct because all squares are similar;

② is incorrect because their angles could have different measures;

③ is incorrect because, despite having the same side lengths, their angle measures could be different;

④ is incorrect because diagonals being equal doesn't matter; what makes rectangles similar is their length-to-width ratio.

Therefore the only correct statement is ①, and thus the correct number is 1.

So the correct answer is 1.

3. According to the following sets of conditions, the number of sets which can be used to deduce that $\triangle ABC$ is similar to $\triangle A_1B_1C_1$ is _____ .

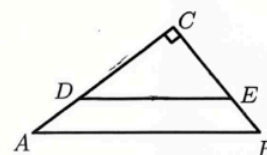
- ① $m\angle A = 45^\circ$, $AB = 12$, $AC = 15$, $m\angle A_1 = 45^\circ$, $A_1B_1 = 16$, $A_1C_1 = 20$;
 ② $AB = 12$, $BC = 15$, $AC = 24$, $A_1B_1 = 20$, $A_1C_1 = 40$, $B_1C_1 = 25$;
 ③ $m\angle B = m\angle B_1 = 75^\circ$, $m\angle C = 50^\circ$, $m\angle A_1 = 55^\circ$;
 ④ $m\angle C = m\angle C_1 = 90^\circ$, $AB = 10$, $AC = 6$, $A_1B_1 = 15$, $A_1C_1 = 9$.

A. 1 B. 2 C. 3 D. 4

3. 【 Answer 】 D

【 Solution 】 ① is correct because two pairs of corresponding sides are in proportion and their included angles have the same measures;
 ② is correct because three pairs of corresponding sides are in proportion;
 ③ is correct because two pairs of corresponding angles are equal;
 ④ by the Pythagorean Theorem, $BC = 8$, and $B_1C_1 = 12$, so all three pairs of sides are all in proportion, therefore ④ is correct.
 Therefore the correct answer is D.

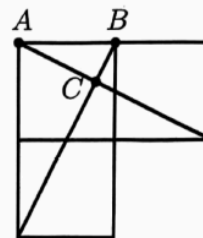
5. As shown in the figure, in $\triangle ABC$, $m\angle C = 90^\circ$. $AC = 4$, $BC = 3$, and $\overline{DE} \parallel \overline{AB}$. If the area of $\triangle CDE$ is equal to the area of $DABE$, calculate the lengths of $\overline{CD}$ and $\overline{DE}$.



5. 【 Answer 】 $CD = 2\sqrt{2}$, $DE = \frac{5}{2}\sqrt{2}$

【 Solution 】 Since the area of $\triangle CDE$ is equal to the area of $DABE$, and $\triangle CDE \sim \triangle CAB$, we have
 $AB = \sqrt{2}DE$. $AB = 5$, $DE = \frac{5}{2}\sqrt{2}$, $CD = 2\sqrt{2}$.

- 7 Three unit squares and two line segments connecting two pairs of vertices are shown. What is the area of $\triangle ABC$? (2012 AMC 10A Problem, Question #15)



A. $\frac{1}{6}$

B. $\frac{1}{5}$

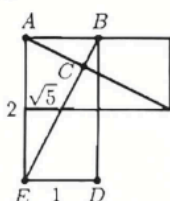
C. $\frac{2}{9}$

D. $\frac{1}{3}$

E. $\frac{\sqrt{2}}{4}$

7. 【 Answer 】 B

【 Solution 】



$\triangle ABC \sim \triangle BED$. The hypotenuse of right triangle BED is $\sqrt{1^2 + 2^2} = \sqrt{5}$.

$$\frac{AC}{BC} = \frac{BD}{ED} \Rightarrow \frac{AC}{BC} = \frac{2}{1} \Rightarrow AC = 2BC,$$

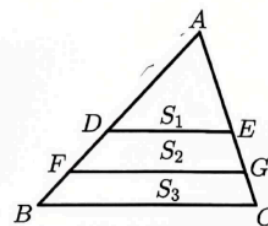
$$\frac{AC}{AB} = \frac{BD}{BE} \Rightarrow \frac{AC}{1} = \frac{2}{\sqrt{5}} \Rightarrow AC = \frac{2}{\sqrt{5}},$$

Since $AC = 2BC$, $BC = \frac{1}{\sqrt{5}}$. $\triangle ABC$ is a right triangle so the area is just

$$\frac{1}{2} \cdot AC \cdot BC = \frac{1}{2} \cdot \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} = \boxed{(B) \frac{1}{5}}.$$

.L11 E

6. As shown in the figure, $\triangle ABC$ is divided into three separate parts by $\overline{DE}$ and $\overline{FG}$, such that $S_1 = 2S_2 = 3S_3$ and that $\overline{DE} \parallel \overline{FG} \parallel \overline{BC}$. Then $DE : FG : BC = \underline{\hspace{2cm}}$.



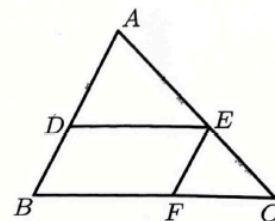
6. 【 Answer 】 $\sqrt{6} : 3 : \sqrt{11}$

【 Solution 】 Since $S_1 = 2S_2 = 3S_3$, the area of triangle AFG is $\frac{3}{2}S_1$, and the area of triangle ABC is $\frac{11}{6}S_1$.

The ratio of the area of three triangles is $1 : \frac{3}{2} : \frac{11}{6}$, which is $6 : 9 : 11$, so the

$$DE : FG : BC = \sqrt{6} : 3 : \sqrt{11}.$$

7. As shown in the figure, in $\triangle ABC$, point D and point E are midpoints of $\overline{AB}$ and $\overline{AC}$, respectively. Draw $\overline{DE}$, then the ratio of the area of $\triangle ADE$ to the area of quadrilateral $BCED$ is $\underline{\hspace{2cm}}$.



7. 【 Answer 】 $1 : 3$

【 Solution 】 Since D and E are, respectively, midpoints of $\overline{AB}$ and $\overline{AC}$,

$$\overline{DE} \parallel \overline{BC}, DE = \frac{1}{2}BC.$$

Therefore $\triangle ADE \sim \triangle ABC$,

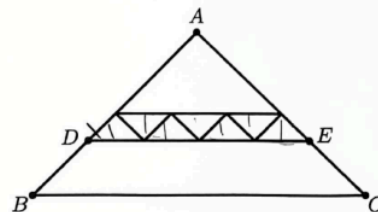
$$\text{so } \frac{S_{\triangle ADE}}{S_{\triangle ABC}} = \left(\frac{DE}{BC}\right)^2 = \frac{1}{4}.$$

$$\text{So } \frac{S_{\triangle ADE}}{S_{BCED}} = \frac{1}{3}.$$

As such, the correct answer is $1 : 3$.

L12

7. All of the triangles in the diagram below are similar to isosceles triangle ABC , in which $AB = AC$. Each of the 7 smallest triangles has area 1, and $\triangle ABC$ has area 40. What is the area of trapezoid $DBCE$? (2018 AMC 10A Problem, Question #9)

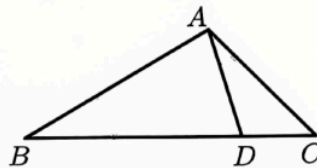


- A. 16 B. 18 C. 20 D. 22 E. 24

1. 【 Answer 】 E

【 Solution 】 Let x be the area of ADE . Note that x is comprised of the 7 small isosceles triangles and a triangle similar to ADE with side length ratio 3 : 4 (so an area ratio of 9 : 16). Thus, we have $x = 7 + \frac{9}{16}x$. This gives $x = 16$, so the area of $DBCE = 40 - x = \boxed{\text{(E) } 24}$.

5. As shown in the figure, in $\triangle ABC$, $AC = 2$, $BC = 4$, and D is a point on $\overline{BC}$. $m\angle CAD = m\angle B$. If $\triangle ADC$ has an area of a , then the area of $\triangle ABD$ is _____.



- A. $2a$ B. $\frac{5}{2}a$ C. $3a$ D. $\frac{7}{2}a$

5. 【 Answer 】 C

【 Solution 】 Since $m\angle CAD = m\angle B$, and $m\angle ACD = m\angle BCA$,
 $\triangle ACD \sim \triangle BCA$,

so we have $\frac{S_{\triangle ACD}}{S_{\triangle BCA}} = \left(\frac{AC}{BC}\right)^2$, which is equivalent to $\frac{a}{S_{\triangle BCA}} = \frac{1}{4}$.

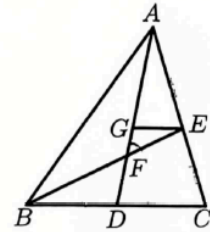
Therefore the area of $\triangle BCA$ is $4a$.

So the area of $\triangle ABD$ is $4a - a = 3a$.

The correct answer, thus, is C.

- 7 As shown in the figure, in $\triangle ABC$, $\overline{AD}$ and $\overline{BE}$ are medians that intersect at point F . $\overline{EG} \parallel \overline{BC}$, intersecting with $\overline{AD}$ at point G .

(1) Prove that $\frac{AG}{AD} = \frac{GF}{FD}$.



【 Solution 】 (1) Since $\overline{EG} \parallel \overline{BC}$,
 $m\angle GEF = m\angle DBF$.
 Since $m\angle GFE = m\angle DFB$,
 $\triangle EGF \sim \triangle BDF$, so $\frac{GF}{DF} = \frac{GE}{BD}$.
 Note that since $\overline{EG} \parallel \overline{BC}$,
 we have $\triangle AGE \sim \triangle ADC$, so $\frac{GE}{DC} = \frac{AG}{AD}$.
 Since D is the midpoint of BC ,
 $BD = CD$.
 Therefore $\frac{GF}{DF} = \frac{GE}{BD} = \frac{GE}{CD} = \frac{AG}{AD}$, so $\frac{AG}{AD} = \frac{GF}{FD}$.

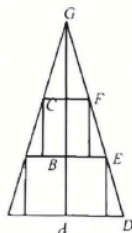
4. A square with side length 3 is inscribed in an isosceles triangle with one side of the square along the base of the triangle. A square with side length 2 has two vertices on the other square and the other two on sides of the triangle, as shown. What is the area of the triangle?(2021 AMC Fall 10B, Question #13)

A. $19\frac{1}{4}$ B. $20\frac{1}{4}$ C. $21\frac{3}{4}$ D. $22\frac{1}{2}$ E. $23\frac{3}{4}$

4. 【 Answer 】 B

【 Solution 】 Solution 1:

Let's split the triangle down the middle and label it:



We see that $\triangle ADG \sim \triangle BEG \sim \triangle CFG$ by AA similarity. $BE = \frac{3}{2}$ because AG cuts the side length of the square in half; similarly, $CF = 1$. Let $CG = h$; then by side ratios,

$$\frac{h+2}{h} = \frac{\frac{3}{2}}{1} \Rightarrow 2(h+2) = 3h \Rightarrow h = 4$$

Now the height of the triangle is $AG = 4 + 2 + 3 = 9$.

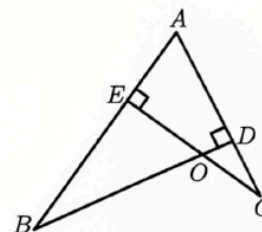
By side ratios, $\frac{9}{4} = \frac{AD}{1} \Rightarrow AD = \frac{9}{4}$ The area of the triangle is $AG \cdot AD = 9 \cdot \frac{9}{4} = \frac{81}{4} = B$

Solution 2:

By similarity, the height is $3 + \frac{3}{1} \cdot 2 = 9$ and the base is $\frac{9}{2} \cdot 1 = 4.5$. Thus the area is

$$\frac{9 \cdot 4.5}{2} = 20.25 = 20\frac{1}{4}, \text{ or (B)}$$

5. As shown in the figure, $\overline{BD} \perp \overline{AC}$ at point D , and $\overline{CE} \perp \overline{AB}$ at point E . $\overline{BD}$ intersects with $\overline{CE}$ at point O . If $AB = 4$, and $AC = 3$, then which of the following statement is incorrect?



- A. $\frac{OC}{OD} = \frac{OB}{OE}$
 B. $\frac{AE}{AD} = \frac{3}{4}$
 C. $\frac{\text{Area of } \triangle ACE}{\text{Area of } \triangle ABD} = \frac{16}{9}$
 D. $\frac{\text{Perimeter of } \triangle ABD}{\text{Perimeter of } \triangle ACE} = \frac{4}{3}$

5. 【 Answer 】 C

【 Solution 】 A: since $\overline{BD} \perp \overline{AC}$, and $\overline{CE} \perp \overline{AB}$,

$$m\angle OEB = 90^\circ, m\angle ODC = 90^\circ.$$

Therefore $m\angle OEB = m\angle ODC$.

Also since $m\angle EOB = m\angle DOC$,

$$\triangle OEB \sim \triangle ODC.$$

$$\text{Therefore } \frac{OE}{OD} = \frac{OB}{OC},$$

$$\text{so } \frac{OC}{OD} = \frac{OB}{OE}.$$

Therefore A is correct;

B: since $\overline{BD} \perp \overline{AC}$, $\overline{CE} \perp \overline{AB}$,

$$m\angle AEC = 90^\circ, \text{ and } m\angle ADB = 90^\circ.$$

Therefore $m\angle AEC = m\angle ADB$.

Also since that $m\angle A = m\angle A$,

$$\triangle AEC \sim \triangle ADB.$$

$$\text{Therefore } \frac{AE}{AD} = \frac{AC}{AB} = \frac{EC}{DB} = \frac{3}{4},$$

so B is correct;

$$\text{C: since } \triangle AEC \sim \triangle ADB, \text{ and } \frac{AC}{AB} = \frac{3}{4},$$

$$\frac{\text{Area of } \triangle ACE}{\text{Area of } \triangle ABD} = \left(\frac{3}{4}\right)^2 = \frac{9}{16},$$

so C is incorrect;

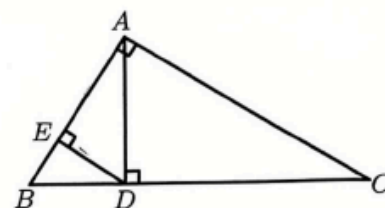
$$\text{D: since } \triangle AEC \sim \triangle ADB, \text{ and } \frac{AC}{AB} = \frac{3}{4},$$

$$\frac{\text{Perimeter of } \triangle ABD}{\text{Perimeter of } \triangle ACE} = \frac{4}{3},$$

so D is correct.

Therefore the correct answer is C.

- 7 In the figure, in right triangle ABC , $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at D , and $\overline{DE} \perp \overline{AB}$ at E . If $AD = 3$ and $DE = 2$, then the lengths of $\overline{AB}$ and $\overline{AC}$ are _____.



- A. $\frac{9\sqrt{5}}{5}, \frac{9}{2}$ B. $\frac{9\sqrt{5}}{5}, \frac{21}{2}$ C. $\frac{9}{5}, \frac{9\sqrt{5}}{2}$ D. $\frac{9}{5}, \frac{4\sqrt{15}}{5}$

7. 【 Answer 】 A

【 Solution 】 Since $AD = 3$ and $DE = 2$, we have $AE = \sqrt{5}$.

Since $AD^2 = AE \cdot AB$, we can find that $AB = \frac{9\sqrt{5}}{5}$.

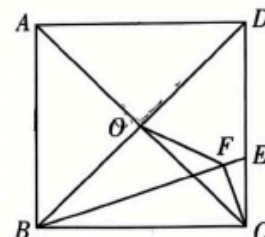
Therefore, $\triangle AED \sim \triangle CAB$.

Hence, $\frac{AE}{AC} = \frac{DE}{AB}$, $\frac{AB}{AC} = \frac{DE}{AE} = \frac{2}{\sqrt{5}}$.

Since $AB = \frac{9\sqrt{5}}{5}$, according to $\frac{AB}{AC} = \frac{2}{\sqrt{5}}$, we can find that $AC = \frac{9}{2}$.

- 8 As shown in the figure, the side length of square $ABCD$ is 6. Point O is the intersection of the diagonals $\overline{AC}$ and $\overline{BD}$. Point E is on the line segment CD . Line segment CF is perpendicular to $\overline{BE}$ at point F , and the foot of the perpendicular is point F . Segment OF is then connected.

(1) Prove that $\triangle BOF \sim \triangle BED$.



(2) If $DE = 2CE$, find OF .

8. 【Answer】 (1) See analysis.

$$(2) OF = \frac{6\sqrt{5}}{5}$$

【Solution】 (1) Since quadrilateral $ABCD$ is a square, we have $\overline{OC} \perp \overline{BO}$, $m\angle BCD = 90^\circ$, we have $BC^2 = BO \cdot BD$.

Since $\overline{CF} \perp \overline{BE}$,

then $BC^2 = BF \cdot BE$,

thus $BO \cdot BD = BF \cdot BE$, which means $\frac{BO}{BE} = \frac{BF}{BD}$, and $m\angle OBF = m\angle EBD$,

hence $\triangle BOF \sim \triangle BED$.

(2) Since $BC = CD = 6$ and $DE = 2CE$,

we have $DE = 4$, $CE = 2$.

In right triangle $\triangle BCE$, $BE = \sqrt{2^2 + 6^2} = 2\sqrt{10}$.

In right triangle $\triangle OBC$, $OB = \frac{\sqrt{2}}{2}BC = 3\sqrt{2}$.

Since $\triangle BOF \sim \triangle BED$,

we have $\frac{OF}{DE} = \frac{BO}{BE}$, which means $\frac{OF}{4} = \frac{3\sqrt{2}}{2\sqrt{10}}$,

thus $OF = \frac{6\sqrt{5}}{5}$.

9. In rectangle $ABCD$, $AB = 6$ and $BC = 3$. Point E between B and C , and point F between E and C are such that $BE = EF = FC$. Segments $\overline{AE}$ and $\overline{AF}$ intersect $\overline{BD}$ at P and Q , respectively. The ratio $BP : PQ : QD$ can be written as $r : s : t$ where the greatest common factor of r , s and t is 1. What is $r + s + t$? (2016 AMC 10A Problem, Question #19)

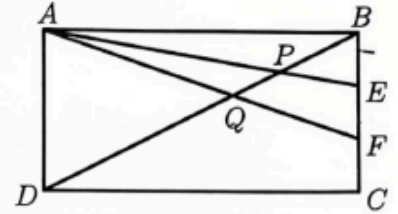
A. 7

B. 9

C. 12

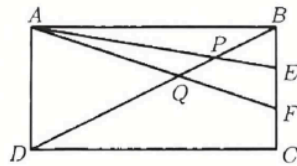
D. 15

E. 20



9. 【 Answer 】 E

【 Solution 】



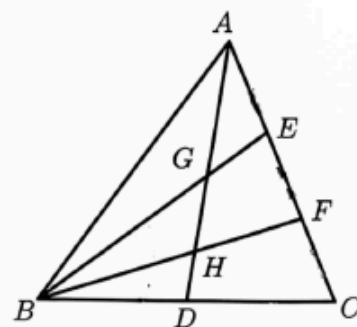
Use similar triangles. Our goal is to put the ratio in terms of $\overline{BD}$.

Since $\triangle APD \sim \triangle EPB$, $\frac{DP}{PB} = \frac{AD}{BE} = 3$. Similarly, $\frac{DQ}{QB} = \frac{3}{2}$. This means that $DQ = \frac{3 \cdot BD}{5}$.

As $\triangle ADP$ and $\triangle BEP$ are similar, we see that $\frac{PD}{PB} = \frac{3}{1}$. Thus $PB = \frac{BD}{4}$.

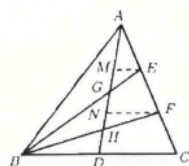
Therefore, $r : s : t = \frac{1}{4} : \frac{2}{5} - \frac{1}{4} : \frac{3}{5} = 5 : 3 : 12$, so $r + s + t = 20$.

10. In the figure, in triangle ABC , point D is the midpoint of side $\overline{BC}$, and points E and F are the trisection points of side $\overline{AC}$. $\overline{AD}$ intersects $\overline{BE}$ and $\overline{BF}$ at points G and H , respectively. Then $AG : GH : HD = \underline{\hspace{2cm}}$.



10. 【 Answer 】 5 : 3 : 2

【 Solution 】



As shown in the figure, line EM is drawn parallel to segment BC passing through point E and intersects line AD at point M . Line FN is drawn parallel to BC passing through point F and intersects line AD at point N .

Since points E and F are the trisection points of segment AC , $AE = EF = FC$,
 $AM = MN = ND$.

Since point D is the midpoint of segment BC , $BD = DC$.

Let $AM = MN = ND = a$,

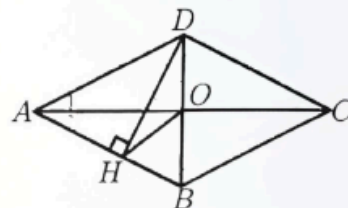
Since $ME = \frac{1}{3}DC = \frac{1}{3}BD$, $\frac{MG}{GD} = \frac{ME}{BD} = \frac{1}{3}$, $MG = \frac{1}{4}MD = \frac{1}{2}a$.

Since $NF = \frac{2}{3}CD = \frac{2}{3}BD$, $\frac{NH}{HD} = \frac{NF}{BD} = \frac{2}{3}$, $HD = \frac{3}{5}ND = \frac{3}{5} \times \frac{1}{3} \times 3a = \frac{3}{5}a$,

$GH = 3a - a - \frac{1}{2}a - \frac{3}{5}a = \frac{9}{10}a$, $AG : GH : HD = \left(a + \frac{1}{2}a\right) : \frac{9}{10}a : \frac{3}{5}a = 5 : 3 : 2$.

L13
Q2

As shown in the figure, $ABCD$ is a rhombus. Its diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point O , $\overline{DH} \perp \overline{AB}$ at point H , and draw $\overline{OH}$. If $m\angle CAD = 20^\circ$, then $m\angle DHO = \underline{\hspace{2cm}}$.



- A. 20° B. 25° C. 30° D. 40°

2. 【 Answer 】 A

【 Solution 】 Since $ABCD$ is a rhombus, we have $OB = OD$ and $\overline{AC} \perp \overline{BD}$.

Since $\overline{DH} \perp \overline{AB}$, we have $OH = OB = \frac{1}{2}BD$.

then $m\angle CAD = 20^\circ = m\angle OAB = m\angle BDH = m\angle OHD$.

3 Which of the following conditions can be used to determine that a quadrilateral is a rhombus?

- A. A pair of opposite sides are parallel and equal, and one angle is a right angle.
- B. One pair of opposite sides are parallel, the other pair of opposite sides are equal, and the two diagonals are perpendicular to each other.
- C. Two pairs of opposite sides are equal respectively, and a pair of adjacent angles are equal.
- D. A pair of opposite sides are equal, a pair of adjacent sides are equal, and a pair of opposite angles are equal.

3. 【 Answer 】 D

【 Solution 】 A is incorrect because this condition is indicative of a rectangle instead of a rhombus;

B is incorrect. B could be representative of a trapezoid whose diagonals are perpendicular to each other;

C is incorrect. From the first condition, this quadrilateral must be a parallelogram. A parallelogram with a set of adjacent angles being equal is a rectangle instead of a rhombus;

D is correct. The proof is as follows:

given that $AB = AD = BC$, $m\angle BAD = m\angle BCD$, draw the figure:

Draw $\overline{AC}$. We know that $AB = BC$,

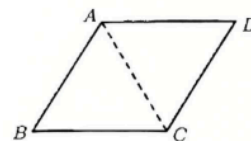
so it must be true that $m\angle BAC = m\angle BCA$,

which means that $m\angle BAD - m\angle BAC = m\angle BCD - m\angle BCA$.

This is equivalent to $m\angle DAC = m\angle DCA$.

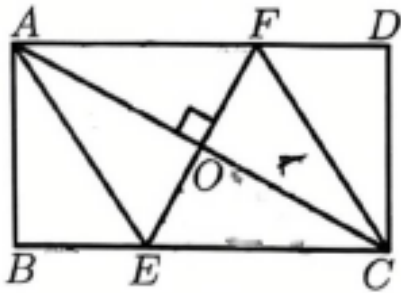
Therefore $DA = DC$, so $AB = BC = DC = DA$,

so $ABCD$ is a rhombus.



- 9 As shown in the figure, in the rectangle $ABCD$, O is the midpoint of the diagonal AC . A line is drawn through point O , intersecting the sides BC and AD at points E and F , respectively. Connect AE and CF .

(2) If $AB = 6$, $BC = 8$, find the side length of rhombus $AECF$.



(2) $\because$ The quadrilateral $ABCD$ is a rectangle,

$\therefore \angle B = 90^\circ$,

$\because$ The quadrilateral $AECF$ is a rhombus,

$\therefore AE = CE$,

Let the side length of the rhombus be x ,

Then $AE = CE = x$, $BE = 8 - x$,

In $\text{Rt}\triangle ABE$ $AB^2 + BE^2 = AE^2$,

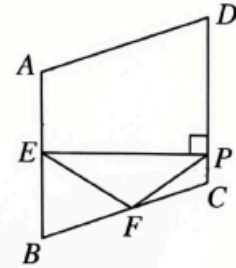
Namely $6^2 + (8 - x)^2 = x^2$,

Solve: $x = \frac{25}{4}$,

So the side length of the rhombus is $\frac{25}{4}$.

L13 E

- 1 As shown in the figure, in rhombus $ABCD$, $m\angle A = 110^\circ$, points E and F are midpoints of $\overline{AB}$ and $\overline{BC}$, respectively. $\overline{EP} \perp \overline{CD}$ at point P . Then $m\angle PEF = \underline{\hspace{1cm}}$.



A. 35°

B. 45°

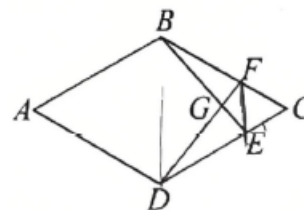
C. 50°

D. 55°

1. 【 Answer 】 A

【 Solution 】 In rhombus $ABCD$, $m\angle A = 110^\circ$,
we have $m\angle B = 70^\circ$ and $BE = BF$.
Therefore $m\angle BEF = m\angle BFE = 55^\circ$.
Note that since $m\angle PEB = 90^\circ$,
we have $m\angle FEP = 35^\circ$.
So the correct answer is A .

5. As shown in the diagram, in the rhombus $ABCD$, E , F are points on sides CD and BC , respectively. $CE = CF$. Connect BE and DF , intersecting at point G . Prove that $BE = DF$.



5. 【 Answer 】 Same as the analysis.

【 Solution 】 $\because$ Quadrilateral $ABCD$ is a rhombus.

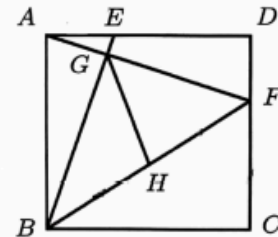
$$\therefore BC = DC.$$

$$\text{In } \triangle CBE \text{ and } \triangle CDF, \begin{cases} BC = DC \\ \angle C = \angle C, \\ CE = CF \end{cases}$$

$$\therefore \triangle CBE \cong \triangle CDF \text{ (SAS),}$$

$$\therefore BE = DF.$$

7. The side length of the square $ABCD$ is 8, points E and F are on AD and CD respectively, $AE = DF = 2$, BE intersects with AF at point G , point H is the midpoint of BF . Connect GH , then the length of GH is _____ .



7. 【 Answer 】 5

【 Solution 】 $\because$ The quadrilateral $ABCD$ is a square,

$$\therefore \angle BAE = \angle D = 90^\circ, AB = AD,$$

$$\therefore AB = AD, \angle BAE = \angle D, AE = DF,$$

$$\therefore \triangle ABE \cong \triangle DAF \text{ (SAS)},$$

$$\therefore \angle ABE = \angle DAF,$$

$$\therefore \angle ABE + \angle BEA = 90^\circ,$$

$$\therefore \angle DAF + \angle BEA = 90^\circ,$$

$$\therefore \angle AGE = \angle BGF = 90^\circ,$$

$$\therefore \text{Point } H \text{ is the midpoint of } BF,$$

$$\therefore GH = \frac{1}{2}BF,$$

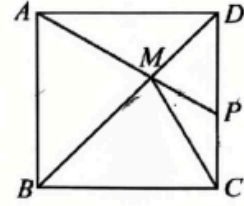
$$\therefore BC = 8, CF = CD - DF = 8 - 2 = 6,$$

$$\therefore BF = \sqrt{BC^2 + CF^2} = 10,$$

$$\therefore GH = 5,$$

Thus the answer is: 5.

- 8 As shown in the figure, in the square $ABCD$ with side length 6, M is a point on the diagonal BD . Connect AM and extend it to intersect CD at point P . If $PM = PC$, what is the length of AM ?



- A. $3(\sqrt{3} - 1)$ B. $3(3\sqrt{3} - 2)$ C. $6(\sqrt{3} - 1)$ D. $6(3\sqrt{3} - 2)$

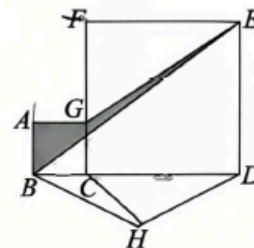
8. 【 Answer 】 C

【 Solution 】 $\because$ The quadrilateral $ABCD$ is a square with a side length of 6. We have
 $\therefore AD = CD = 6, \angle ADC = 90^\circ, \angle ADM = \angle CDM = 45^\circ$. In $\triangle ADM$ and $\triangle CDM$,

$$\begin{cases} DM = DM \\ \angle ADM = \angle CDM = 45^\circ \\ AD = CD \end{cases}; \therefore \triangle ADM \cong \triangle CDM \text{ (SAS)}; \therefore \angle DAM = \angle DCM; \therefore PM = PC;$$

 $\therefore \angle CMP = \angle DCM; \therefore \angle APD = \angle CMP + \angle DCM = 2\angle DCM = 2\angle DAM$. Moreover,
 $\therefore \angle APD + \angle DAM = 180^\circ - \angle ADC = 90^\circ$. Assume $PD = x$, then $AP = 2PD = 2x$;
 $PM = PC = CD - PD = 6 - x; \therefore AD = \sqrt{AP^2 - PD^2} = \sqrt{3}x = 6$. Solving gives $x = 2\sqrt{3}$;
 $\therefore PM = 6 - x = 6 - 2\sqrt{3}; AP = 2x = 4\sqrt{3}$;
 $\therefore AM = AP - PM = 4\sqrt{3} - (6 - 2\sqrt{3}) = 6(\sqrt{3} - 1)$. Therefore, the answer is: C.

9. As shown in the figure, in square $ABCG$ and square $CDEF$, point G is the trisection point of CF , point H and point A are centrosymmetric to point C . Connect EG , BE , BH , DH . If the shaded area is 4, then the length of BH is _____ .



9. 【 Answer 】 $2\sqrt{5}$

【 Solution 】 Let the side length of the square $ABCG$ be a .

$\because$ point G is the trisection point of CF ,

$\therefore CF = 3CG$,

The side length of square $CDEF$ is $3a$,

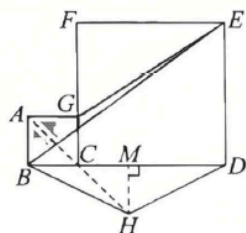
$\therefore BD = BC + CD = 4a$, $FG = FC - CG = 2a$,

$\therefore A_{\text{shaded}} = A_{ABCG} + A_{FCDE} - A_{\triangle FGE} - A_{\triangle BDE} = 4$,

thus the equation can be obtained as $a^2 + 9a^2 - 3a^2 - 6a^2 = 4$,

solving it gives $a = 2$ (negative value discarded),

As shown in the figure, connect AC , CH , passing through point H to make $HM \perp BD$ intersect at point M ,



$\because$ point H and point A are centrally symmetric to point C ,

$\therefore A, C, H$ three points collinear, and $AC = CH$,

$\therefore \angle ABC = \angle HMC$, $\angle ACB = \angle HCM$,

$\therefore \triangle ACB \cong \triangle HCM$ (AAS),

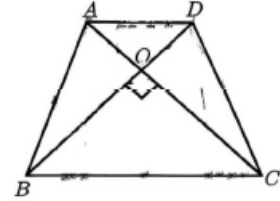
$\therefore MH = CM = AC = BC = 2$,

$\therefore BM = 4$,

$\therefore BH = \sqrt{BM^2 + MH^2} = 2\sqrt{5}$,

Thus the answer is: $2\sqrt{5}$.

2. As shown in the figure, in isosceles trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$. $AB = DC$, $\overline{AC} \perp \overline{BD}$. If $AD = 4$ and $BC = 6$, then the area of $ABCD$ is _____.



2. 【 Answer 】 25

【 Solution 】 Draw $\overline{DE} \parallel \overline{AC}$ through D , intersecting with the extension of $\overline{BC}$ at point E .

Since $\overline{AD} \parallel \overline{BC}$,

$ACED$ is a parallelogram,

so $AC = DE$, $CE = AD = 4$.

Therefore $BE = BC + CE = 6 + 4 = 10$.

Since $\overline{AC} \perp \overline{BD}$,

$\overline{DE} \perp \overline{BD}$.

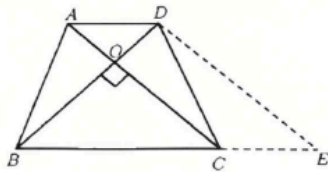
Since $ABCD$ is an isosceles trapezoid,

$AC = BD$,

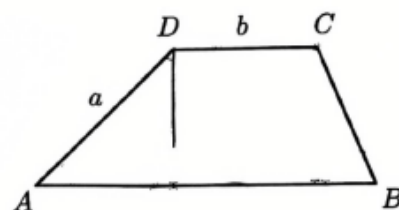
so $BD = DE$,

so we have $BD = DE = \frac{BE}{\sqrt{2}} = 5\sqrt{2}$.

Therefore $S_{ABCD} = \frac{1}{2} \times AC \times BD = 25$.



3. In quadrilateral $ABCD$, it is given that $\overline{AB} \parallel \overline{CD}$ and that $m\angle D = 2m\angle B$. Lengths of $\overline{AD}$ and $\overline{CD}$ are equal to a and b , respectively. Then $AB = \underline{\hspace{2cm}}$.



3. 【 Answer 】 $a + b$

【 Solution 】 Draw $\overline{CE} \parallel \overline{AD}$ through C such that it intersects with $\overline{AB}$ at point E .

Since $\overline{AB} \parallel \overline{CD}$,
quadrilateral $AECD$ is a parallelogram.

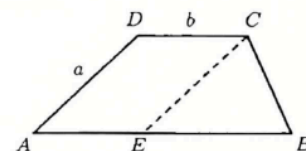
Therefore $AE = CD = b$, $EC = AD = a$, and $m\angle AEC = m\angle D$.

Note that $m\angle D = 2m\angle B$ and that $m\angle CEA = m\angle ECB + m\angle B$;
we have $m\angle ECB = m\angle B$.

Therefore $EC = EB = a$.

So $AB = AE + EB = a + b$.

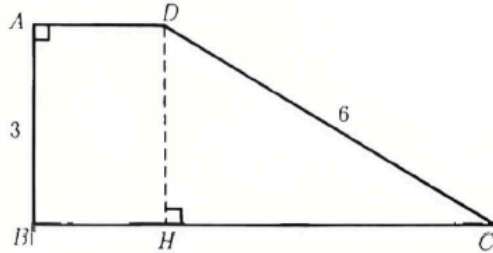
So the correct answer is $a + b$.



6. In the right trapezoid $ABCD$, $AD \parallel BC$, $\angle A = 90^\circ$, $AB = 3$, $CD = 6$, then the measure of $\angle D$ is _____.

6. 【 Answer 】 30° or 150°

【 Solution 】 ①As shown in the figure,

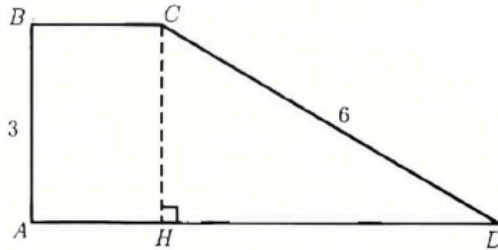


Through point D , draw a line $DH \perp BC$ intersecting at point H ,

then $HC = \sqrt{6^2 - 3^2} = 3\sqrt{3}$,

$\therefore$ at this moment $\angle C = 30^\circ$, $\angle ADC = 150^\circ$.

②As shown in the figure,

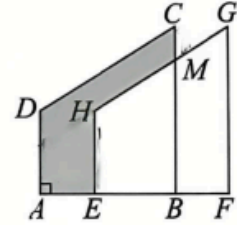


Through point C , draw a line $CH \perp AD$ intersecting at point H ,

then $HD = \sqrt{6^2 - 3^2} = 3\sqrt{3}$, $\angle D = 30^\circ$,

$\therefore \angle D = 30^\circ$ or $\angle D = 150^\circ$.

7. As shown in the figure, in the right trapezoid $ABCD$, $AD \parallel BC$, $AD \perp AB$, $BC = 5$, the right trapezoid $ABCD$ is translated 2 units along the AB direction to obtain the right trapezoid $EFGH$, HG intersects BC at point M , and $CM = 1$. Then, the area of the shaded part in the figure is _____.



7. 【 Answer 】 9

【 Solution 】 $\because$ The right trapezoid $EFGH$ is formed by translating the right trapezoid $ABCD$,
 $\therefore BC = GF = 5$,
 $\because$ The right trapezoid $ABCD$ is translated 2 units along the AB direction to obtain the right trapezoid $EFGH$, and $CM = 1$,
 $\therefore BM = BC - CM = 5 - 1 = 4$, $BF = 2$,
 $\therefore A_{\text{shaded}} + A_{BEHM} = A_{BEHM} + A_{BFGM}$,
 $\therefore A_{\text{shaded}} = A_{BFGM} = \frac{1}{2}(BM + GF) \cdot BF = \frac{1}{2} \times (4 + 5) \times 2 = 9$.
 Thus, the answer is: 9.

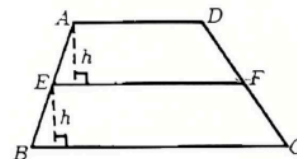
2. It is given that the ratio of the lengths of the upper and lower bases of a trapezoid is 1 : 2. The median divides the trapezoid into two parts. The ratio of the area of the upper part to that of the lower part is _____ .

2. 【 Answer 】 5 : 7

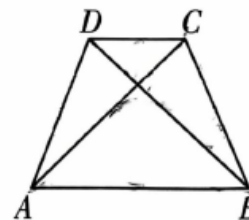
【 Solution 】 Let $AD = 2x$, then $BC = 4x$.

Then $EF = \frac{1}{2}(AD + BC) = 3x$.

$$\frac{S_{AEFD}}{S_{BCFE}} = \frac{\frac{1}{2} \cdot (AD + EF) \cdot h}{\frac{1}{2} \cdot (EF + BC) \cdot h} = \frac{\frac{1}{2} (2x + 3x) \cdot h}{\frac{1}{2} (3x + 4x) \cdot h} = \frac{5}{7}.$$



3. As shown in the figure, in trapezoid $ABCD$, $\overline{AB} \parallel \overline{CD}$, $AD = BC$, diagonals $\overline{AC} \perp \overline{BD}$, and $AC = 5\sqrt{2}$. Then the median of $ABCD$ has a length of _____ .



3. 【 Answer 】 5

【 Solution 】 Draw $\overline{CE} \parallel \overline{BD}$ through C , intersecting with the extension of $\overline{AB}$ at point E .

Since $\overline{AB} \parallel \overline{CD}$, $\overline{CE} \parallel \overline{BD}$,

$DBEC$ is a parallelogram.

Therefore $CE = BD$, $BE = CD$.

Since in isosceles trapezoid $ABCD$, $AC = BD$,

$CE = AC$.

Since $\overline{AC} \perp \overline{BD}$, $\overline{CE} \parallel \overline{BD}$,

we have $\overline{CE} \perp \overline{AC}$.

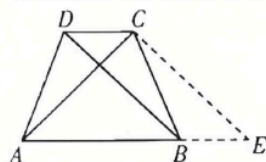
Therefore $\triangle ACE$ is an isosceles right triangle.

Since $AC = 5\sqrt{2}$,

we have $AE = AB + BE = AB + CD = \sqrt{2}AC = 10$.

So the median has a length $= \frac{1}{2}AE = 5$.

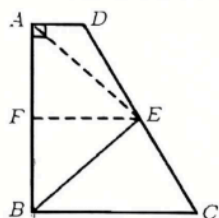
Therefore the answer is 5.



6. In the right trapezoid $ABCD$, $AD \parallel BC$, $AB \perp BC$, E is the midpoint of side CD , and satisfies $AB = AD + BC$, $BE = 3$. What is the area of trapezoid $ABCD$?

6. 【 Answer 】 9

【 Solution 】 Connect AE , draw a line through E intersecting $EF \parallel BC$ at point F ,



$\because E$ is the midpoint of CD ,

$\therefore EF$ bisects AB , EF is the midline of the trapezoid $ABCD$,

therefore $EF = \frac{1}{2}(AD + BC)$,

Also, $\because BC \perp AB$,

$\therefore EF$ is the perpendicular bisector of AB , based on the properties of the perpendicular bisector:

$AE = BE = 3$,

$\because AB = AD + BC$, $EF = \frac{1}{2}(AD + BC) = \frac{1}{2}AB$,

$\therefore \triangle ABE$ is an isosceles right triangle.

Using the Pythagorean theorem: $AB = \sqrt{AE^2 + BE^2} = \sqrt{3^2 + 3^2} = 3\sqrt{2}$,

which is equivalent to $AD + BC = 3\sqrt{2}$,

$$A_{ABCD} = \frac{1}{2}(AD + BC) \cdot AB$$

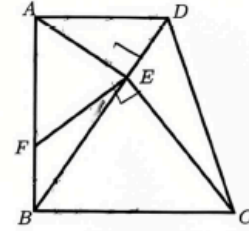
$$= \frac{1}{2}(AD + BC) \cdot (AD + BC)$$

$$= \frac{1}{2} \times 3\sqrt{2} \times 3\sqrt{2}$$

$$= 9.$$

- 7 As shown in the figure, in the right trapezoid $ABCD$, $AD \parallel BC$, $\angle ABC = 90^\circ$, $AE \perp BD$, the foot of the perpendicular is E . Connect CE , and draw $EF \perp CE$, intersecting side AB at point F .

(1) Prove that: $\triangle AEF \sim \triangle BEC$.



(2) If $AB = BC$, prove that: $AF = AD$.

【Solution】 (1) $\because AE \perp BD$, $\angle ABC = 90^\circ$, $EF \perp CE$,
 $\therefore \angle AEB = \angle ABC = \angle FEC = 90^\circ$,
 $\therefore \angle EAF + \angle ABE = 180^\circ - \angle AEB = 90^\circ$,
 $\angle EBC + \angle ABE = \angle ABC = 90^\circ$,
 $\therefore \angle EAF = \angle EBC$,
 furthermore: $\because \angle AEF + \angle FEB = \angle AEB = 90^\circ$,
 $\angle BEC + \angle FEB = \angle FEC = 90^\circ$,
 $\therefore \angle AEF = \angle BEC$,
 $\therefore$ between $\triangle AEF$ and $\triangle BEC$,

$$\begin{cases} \angle EAF = \angle EBC \\ \angle AEF = \angle BEC \end{cases}$$

 $\therefore \triangle AEF \sim \triangle BEC$.

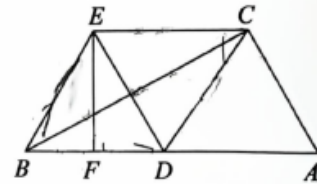
(2) $\because$ From (1) we know that $\triangle AEF \sim \triangle BEC$,
 $\therefore \frac{AE}{BE} = \frac{AF}{BC}$,
 furthermore: $AB = BC$,
 $\therefore \frac{AE}{BE} = \frac{AF}{AB}$,
 $\because AD \parallel BC$, $\angle ABC = 90^\circ$,
 $\therefore \angle BAD = 180^\circ - \angle ABC = 90^\circ$,
 furthermore: $AE \perp BE$,
 $\therefore \angle AEB = \angle DEA = 90^\circ$,
 furthermore: $\because \angle BAE + \angle DAE = \angle BAD = 90^\circ$,
 $\angle BAE + \angle ABE = 180^\circ - \angle AEB = 90^\circ$,
 $\therefore \angle ABE = \angle DAE$,
 $\therefore$ between $\triangle ABE$ and $\triangle DAE$,

$$\begin{cases} \angle AEB = \angle DEA \\ \angle ABE = \angle DAE \end{cases}$$

 $\therefore \triangle ABE \sim \triangle DAE$,
 $\therefore \frac{BE}{AE} = \frac{AB}{AD}$,
 $\therefore \frac{AE}{BE} = \frac{AD}{AB}$,
 furthermore: $\frac{AE}{BE} = \frac{AF}{AB}$,
 $\therefore AF = AD$.

- 8 As shown in the figure, in $\text{Rt}\triangle ABC$, $m\angle ACB = 90^\circ$. Point D is on $\overline{AB}$. Draw $\overline{CD}$, $\overline{CE} \parallel \overline{AB}$, and $\overline{BE} \parallel \overline{CD}$, such that $CE = AD$.

(1) Prove that quadrilateral $BDCE$ is a rhombus.



(2) Draw $\overline{EF} \perp \overline{BD}$ through point E such that the perpendicular foot is located at F . If F is the midpoint of $\overline{BD}$, and $EB = 6$, calculate the length of $\overline{BC}$.

【 Solution 】 (1) Since $\overline{CE} \parallel \overline{AB}$ and $\overline{BE} \parallel \overline{CD}$,
 $BDCE$ is a parallelogram.
 Since $CE = BD$,
 $CE = AD$,
 so $BD = AD$,
 so we have point D being the midpoint of $\overline{AB}$.
 Since $m\angle ACB = 90^\circ$,

we have $CD = BD = AD$.

Therefore $BDCE$ is a rhombus.

(2) Since $\overline{EF} \perp \overline{BD}$ and F is the midpoint of $\overline{BD}$,
 we have $BE = ED$ and $BE = BD = 6$.

Therefore $m\angle EBD = 60^\circ$ and $m\angle CBA = 30^\circ$,
 so $AC = 6$ and $AB = 2BD = 12$.

As such,

$$BC = \sqrt{AB^2 - AC^2} = \sqrt{12^2 - 6^2} = 6\sqrt{3}.$$